

SE3050 – Deep Learning**LAB 07 – Autoencoders - Report****1. Overview**

This lab focused on autoencoders using the Fashion-MNIST dataset. Three implementations were evaluated: a dense feed-forward autoencoder, a vanilla CNN autoencoder, and a CNN-based image denoising autoencoder. Each model was trained for 30 epochs, evaluated using Mean Squared Error (MSE), and analysed using training and validation loss.

2. Experimental Results

Model	Test MSE	Observation
FFNN Autoencoder	0.008615	Basic reconstruction using dense layers.
Vanilla CNN Autoencoder	0.001808	Lowest reconstruction error among the tested models.
CNN Denoising Autoencoder	0.006501	Reconstructs clean images from noisy inputs.

3. Linear Autoencoder and PCA

A linear autoencoder is an autoencoder that does not use non-linear activation functions. When a linear autoencoder is trained using Mean Squared Error, it learns a lower-dimensional representation that is closely related to Principal Component Analysis (PCA). Both methods aim to reduce dimensionality while preserving the most important information in the data. PCA finds principal components mathematically, whereas a linear autoencoder learns the representation through neural-network training. Therefore, a linear autoencoder can learn a subspace similar to the one obtained by PCA

4. Performance Improvement: FFNN AE vs Vanilla CNN AE

The Vanilla CNN Autoencoder performed better than the FFNN Autoencoder in this experiment. The FFNN Autoencoder produced a test MSE of approximately 0.008615, while the Vanilla CNN Autoencoder produced a much lower test MSE of approximately 0.001808.

The main reason is that convolutional layers preserve and learn spatial information such as local patterns, edges, shapes, and neighbouring pixel relationships. In contrast, the FFNN

Autoencoder flattens the image before processing it, which removes the original two-dimensional spatial arrangement. Therefore, the CNN Autoencoder can reconstruct Fashion-MNIST images more accurately.

5. Image Denoising Autoencoder and Noise Factor

A noise factor of 0.2 was used in the final implementation. This value adds clearly visible noise to the Fashion-MNIST images while keeping the main structure of each object recognizable. This provides a suitable learning problem for the denoising autoencoder, because the model must recover the clean image from a corrupted input.

After 30 epochs, the denoising autoencoder achieved a test MSE of approximately 0.006501. The reconstructed outputs were visibly cleaner than the noisy input images, showing that the model successfully learned to remove a significant amount of noise.

6. Denoising AE vs Vanilla CNN AE

The Vanilla CNN Autoencoder achieved a lower test MSE of approximately 0.001808, whereas the Denoising Autoencoder achieved approximately 0.006501. However, these values should be interpreted together with the task performed by each model. The Vanilla CNN Autoencoder reconstructs an image from a clean input, while the Denoising Autoencoder receives a noisy image and attempts to reconstruct the original clean image.

The denoising model therefore solves a more difficult reconstruction problem. By learning to recover clean images from corrupted inputs, it becomes more robust to noise and is encouraged to learn useful image structure rather than simply copying the input.

7. Difference Between AE and VAE

A **standard Autoencoder (AE)** learns a compressed latent representation of an input and then reconstructs the original data from that representation. Its main objective is usually accurate reconstruction.

A **Variational Autoencoder (VAE)** also learns a latent representation, but instead of mapping each input to a single fixed latent vector, it learns a probability distribution in the latent space, commonly represented using a mean and variance. Samples can then be drawn from this learned distribution.

Because of this probabilistic latent space, VAEs can generate new samples that are similar to the training data. Therefore, standard AEs are commonly used for reconstruction, dimensionality reduction, and denoising, while VAEs are additionally suitable for generative tasks.

8. Conclusion

The lab demonstrated how different autoencoder architectures affect image reconstruction. The Vanilla CNN Autoencoder achieved the lowest reconstruction error because convolutional layers preserve spatial information. The Denoising Autoencoder showed that an autoencoder can also learn to reconstruct clean images from noisy inputs.

The experiments also highlighted the conceptual relationship between linear autoencoders and PCA, and the key difference between deterministic autoencoders and probabilistic Variational Autoencoders.